System Subsystem Design Description

for the

The Balance Project

DCN: SDP20231224-P1:128

Revision Date: 06 Dec, 2025

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Controlled by: AGENCY
Controlled by: OFFICE
CUI Category: WILL-BE
Limited Dissemination Control: TBD
POC: Undefined Gpl, undefined.l.gpl.civ@us.navy.mil

Distribution Statement D: Vinay Agarwal

DOCUMENT CHANGE HISTORY

The following table is a simple list of released revisions sent for review. Records of reviews and the review artifacts are saved with reviewer information in the The Balance Projectartifact repository.

Change Record

Date	Version	Author(s)	Change Reference
13 Sep 2025	P1	Vinay Agarwal	Preliminary DRAFT version

Draft P1 Preliminary version of this document.

1. Baseline Document draft 1

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CHAPTER 1

Scope

This document provides the System / Subsystem Design Description (**SSDD**) for the Balance System. The system will be referred to as the Balance System.

1.1 Identification

The Balance System described in this document shall be known as Balance System version 1. However, the Operational Concept Description **OCD** described herein shall be applicable to pre-releases such as Beta-releases for a phased release as listed for each requirement. The major system interfaces and capabilities are fully specified in Chapter 3.

1.2 System Overview

Figure 1 shows the high-level architecture for the Balance System system. This diagram shows the major external interfaces that provide the capabilities of Balance System.

The Balance System is a single player game. The game is a standalone game played on the STM32 board. This system would be a game where the user would have to balance a ball on a LCD screen that is builtin on the STM32 board. The objective of the game is to balance the ball on the screen based on the way the board was tilted. Using an onboard gyroscope that communicates across the board to the MCU though SPI communication. Balance System would keep track of the current position of the ball and where the next updated move is. This helps keep track of the system of where the ball is until a movement has occurred. Balance System shall process at a maximum 180 Hz. This would give the user enough time to process the current angle of the ball and be able to present on the LCD-TFT screen.

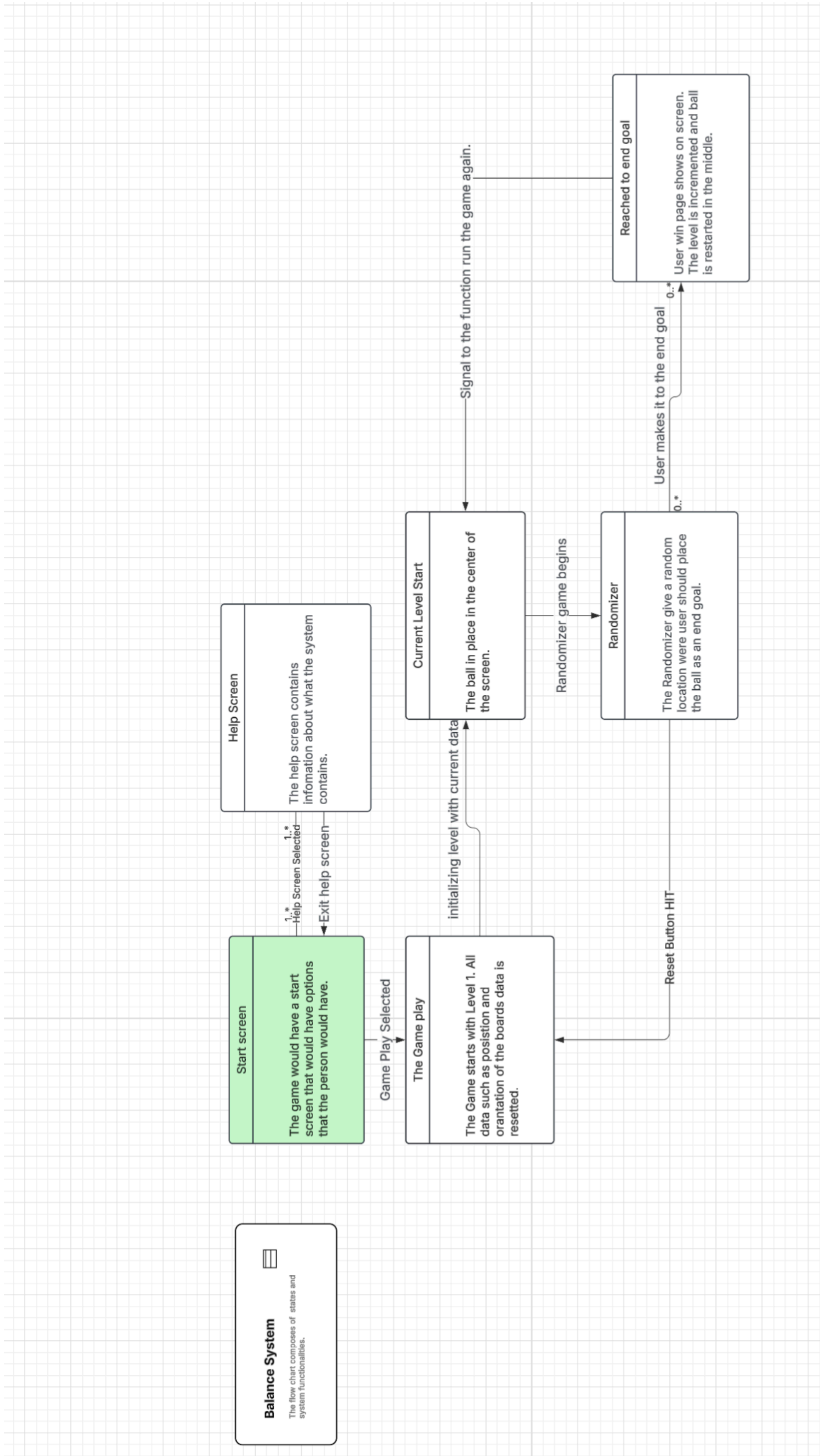

Figure 1: System Overview

1.3 Document Overview

This section provides information about this document's security/privacy considerations, contents, structure, and version information. This section also provides information regarding how specifications are formatted in this artifact and how they can best be understood.

1.3.1 Security and Privacy Considerations

This document has been identified as ”Controlled Unclassified Information” (CUI). Please follow the control block on the title page for ownership, creation, category, dissemination, and Point of Contact (**POC**) information. This information should be delineated, per `https://www.dodcui.mil` as:

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1.3.2 Artifact Format

This document format is based upon the guidance in the **SSDD DID** [**ref˙˙SSDD˙DID**]. The system design is documented following the guidelines of ISO-12207 [**ref˙˙ISO˙12207**] and MIL-STD-498 [**ref˙˙MIL˙STD˙498**] (from which ISO-12207 originated). The goal of this artifact is to provide a road map to the design implementation through either linkage to the development items (for example code, hardware, and **CAD** drawings) or, for larger systems, segregated documentation such as Software Design Description (**SDD**), Hardware Design Description (**HDD**), or Interface Design Description (**IDD**) artifacts. This document follows the listed **SSDD** sub-section order.

Section 1 provides an overview of the system and this document.

Section 2 lists general and application-specific reference documents as well as glossary terms and acronyms.

Section 3 presents the system-wide design decisions.

Section 4 provides the detailed system architecture design.

Section 5 provides any applicable requirement traceability.

Appendices if needed, provide additional information as may be needed.

1.3.3 Document Version Information

This document was produced in LaTeX and *BibLaTeX/Biber*. The editing and document preparation were performed using MiKTeX version 2.9 with the build option [LaTeX ⇒ PS ⇒ PDF]. The LaTeX*svn-multi* package was used to glean SVN tracking information, when files are stored in an “SVN” version control system. The style `KNEADdocument` was used to provide the LaTeX and *BibLaTeX/Biber* formatting details.

This revision of this document has the following properties:

Tracking Item	**Data**
Repository	
Author	KneadProject
Revision	142
Rev Date	2025-08-03 20:42:20Z
Print Date	06 Dec, 2025 15:35
KNEADdocument Version	1.02
KNEADdocument Date	2024/12/21

CHAPTER 2

References

This section provides a list of referenced items for this document.

2.1 Acronyms and Abbreviations

This section defines acronyms and abbreviations used in this and related documents.

Table 1: Acronym Definitions

Acronym	**Definition**
YSM	Yourdon Structured Method
LTDC	Layered Transmission Display Controller
STM	STMicroelectronics
MEMS	Micro-Electro-Mechanical Systems
End of acronym definition table	

2.2 Glossary and Definitions

This section defines glossary terms used in this and related documents.

Table 2: Glossary Terms and Definitions

Glossary Term	**Definition**
STM32F429I	Micro-controller board has all component fit onto one board.
Customer	The professor that is view the grading all assignments.
End of glossary terms table	

2.3 Referenced Documents

This section lists the referenced documents for this document. The references are categorized into two categories:

External Documents not directly associated with this project.

Project Documents that are directly associated with this project.

2.3.1 External Documents

2.3.2 Project Specific Documents

CHAPTER 3

System-wide Design Decisions

The system will always start at the initial state. The initial state would run through all initialize steps to do reading for the system. That means that the Gyroscode will state doing reading and the data structure will be stored in memory. Once the initialization begins then the user should be able to view the main screen that is used for the system. This Main screen is where the user can select 2 buttons on the screen. The user would have 2 options to be able to select the buttons on the main screen. One being touch screen and the other with a scroll button. Both options are located on the board. The Main screen is where the user would select to start the game or read the helpful guide. If the user select the help screen, then a new page would appear. This new page will descriptions of how to play the game. The page will have symbol descriptions and what the users' should be looking for while playing the game. To return to the main screen the users should click the 'X' on the top right. If the user selects the run game button, the game would begin. When the game begins from the main page it always starts at level one. Reach levels page would have the same setup. This setup allows the user how much time they have left at the level and the level there are on. This allows the user to see what then current state of the level they are play that they are at. Every single level that the user is playing the game the ball is place in a random location. The goal where the user should place the ball whould also be placed in a random location. This allows every level to be unique. If the user does not meet the goal at the end of the round then then user losses the game and is taken back to main menu. If the user moves the ball into the goal line then the level is complete and the game is continued.

CHAPTER 4

System Architectural Design

This chapter describes the system architectural design.

4.1 System Components

The System components compose of the following: GYRO scope : reads angular velocities LTDC : screen and touch screen to display angles. SDRAM : hold memory displaying and updating the screen.

4.2 Concept of Execution

First reads the the gyroscope value. Calculates what angles the ball should rotate to. Determain if there is match with the posistion of the ball and the goal.

4.3 Interface Design

3 external interfaces. Gyroscope, SDRAM, and LTDC screen. Then are all interfacing with the STM32ZIT chip on the board.

4.3.1 Interface One

The Gyroscope Interface. Add more descriptions

4.3.2 Interface Two

SDRAM Interface Add more descriptions

4.3.3 Interface Three

LTDC Interface Add more descriptions

CHAPTER 5

Traceability

The systems traceability will be in the order of OCD to SDP to SSS. Each requirement would be traced back from one to the other based on there numbering.

CHAPTER 6

Schedule

The Schedule is based on the time line of the professor.

APPENDIX

Notes

This section provides notes, as necessary, to document the system /subsystem design description.

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